
THE EFFECT OF PLYOMETRIC TRAINING ON POWER AND KICKING DISTANCE IN FEMALE ADOLESCENT SOCCER PLAYERS

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ABSTRACT

Rubley, MD, Haase, AC, Holcomb, WR, Girouard, TJ, and Tandy, RD. The effect of plyometric training on power and kicking distance in female adolescent soccer players. *J Strength Cond Res* 25(1): 129–134, 2011—The purpose of this study was to measure the effects of low-frequency, low-impact plyometric training on vertical jump (VJ) and kicking distance in female adolescent soccer players. Sixteen adolescent soccer players were studied (age 13.4 ± 0.5 years) across 14 weeks. The control group (general soccer training only) had 6 subjects, and the plyometric training (general soccer training plus plyometric exercise) group had 10 subjects. All subjects were tested for VJ and kicking distance on 3 occasions: pre-test, 7 weeks, and 14 weeks. Data were analyzed using a 2 (Training) $\times$ 3 (Test) analysis of variance (ANOVA) with repeated measures on the factor test. No significant difference in kicking distance was found between groups at pre-test ($p = 0.688$) or 7 weeks ($p = 0.117$). The plyometric group had significantly greater kicking distance after 14 weeks ($p < 0.001$). No significant difference in VJ height was found between groups at pre-test ($p = 0.837$) or 7 weeks ($p = 0.108$). The plyometric group had a significantly higher VJ after 14 weeks ($p = 0.014$). These results provide strength coaches with a safe and effective alternative to high-intensity plyometric training. Based on these findings, to increase lower-body power resulting in increased VJ and kicking distance, strength coaches should implement once-weekly, low-impact plyometric training programs with their adolescent athletes.

KEY WORDS vertical jump height, soccer kicking distance, power

INTRODUCTION

Plyometric jump training is characterized by a series of exercises involving hops and jumps used to capitalize on the stretch shortening cycle of the muscle (9). It is believed that training with plyometrics facilitates adaptations in muscle function that will increase an athlete's explosive power, which is defined as force times distance over time (5). Plyometrics has been shown to be an effective tool in increasing lower-body power, as measured by vertical jump (VJ), in several studies using adult subjects (1,11,13,14,17,23). However, despite the increasing popularity of plyometric training with young athletes (32), there is very little empirical evidence regarding the efficacy of plyometric training with adolescents. Furthermore, the bulk of research investigating plyometric training efficacy has looked at high-impact plyometric exercises such as depth jumps (1,8,11,13,14,17,18,23), which may be contraindicated for youth as a result of the high risk of injury to the growth plates possibly resulting in leg-length discrepancy (2,3,12,31,34).

To participate in high-impact plyometric training, it has been suggested that athletes should be able to squat 1.5 to 2 times their body weight (5). If followed, this standard would prevent most adolescent female athletes from ever participating in plyometric training. We believe that low-impact plyometric exercises, which are comparable to typical activities of adolescent athletes, simply structured into an organized regimen can be both safe and effective. Because the age and development of our selected test population, low-impact, once-weekly plyometric training was chosen for this study. The primary aim of this investigation was to test the effectiveness of low-impact and low-frequency plyometric training in adolescent female soccer players. Because plyometric training has been shown to increase power, performance variables that require power were chosen for this study. VJ has been established as an accurate measure of lower-body power (15,16,21), and kicking distance is a specific skill used in soccer. We hypothesize that 12 weeks of low-impact plyometric training on 1 day per week will significantly increase VJ and kicking distance in female

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adolescent soccer athletes. Additionally, this study will provide normative performance values because there is currently very little information available for female adolescent soccer athletes.

METHODS

Experimental Approach to the Problem

A group-assigned, mixed-model experimental design was used to determine the effect of low-impact, low-frequency plyometric training on power-related performance variables. The 2 dependent variables were VJ and kicking distance. The independent variable was type of training. Subjects for the 2 experimental groups participated on 2 relatively equal club teams participating in the same league with nearly equal records; therefore, we assumed the participants were nearly equal in ability, experience, and training level. One team completed only regular soccer practices, thus serving as the control, whereas the other team also completed once-weekly plyometric training, thus serving as the experimental group. The effectiveness of the low-impact plyometric training program on kicking distance and VJ was evaluated with a pre-test, 7 week test, and post-test during week 14.

Subjects

Sixteen female adolescent soccer players (age = 13.4 ± 0.5 year, height = 162.5 ± 5.67 cm, weight = 50.84 ± 5.1 kg) from the local girls soccer league volunteered for this study. The methods of the study were approved by the University's Institutional Review Board for use of human subjects. Subjects and their parents were informed of the purpose of this study and the potential risk of participating in the study. Because subjects were younger than 18 years of age, each subject signed an assent form and a parent signed an informed consent approved by the Office for the Protection of Research Subjects. All subjects were currently participating in customary training for soccer during the fall club season, and the study concluded prior to their next club season starting in the spring.

Subjects were excluded from participation in this study if they had previously engaged in a formal plyometric training program, had fewer than 4 years of club soccer experience, had an orthopedic injury in the past 6 months, or had incurred an injury during the progress of training. During the course of the study, 4 girls were excluded from the control group based on their absence from 1 or more testing sessions.

Procedures

One week prior to testing, subjects were familiarized with the testing protocol, which included warm-up, VJ, and kicking. Prior to all testing subjects warmed up by completing the following dynamic movements over a 50-yard distance: skipping, high knee running, glut (gluteal) kicks, high skipping, side shuffling, and skip/kicks (29). Subjects were tested using their preferred striking leg to jump with and strike the ball during testing on 3 occasions: (a) pre-test, (b) during week 7, and (c) during week 14. Immediately prior to each testing

session, the wind speed was measured using a Gill Wind Gauge (Gill Athletics, Urbana, Illinois, USA) and no testing was completed if the wind speed exceeded 2 mph.

VJ was measured using the Vertec (Sports Imports, Columbus, Ohio, USA) (19), which has been established as a reliable method for testing VJ, which has been linked as an effective measurement of power (15,16,21). Subjects stood directly under the Vertec, fully extending an arm to touch the highest vane possible while remaining flat-footed to establish standing reach height, which was recorded. Subjects were instructed to take 1 step back with their dominant leg and prepare to jump. When prepared, subjects took a 1-step approach into a 2-footed jump and aimed to touch the highest vane possible (4,19,21). Subjects were allowed 1 familiarization jump with the Vertec prior to the actual test (4,19,21). Five test jumps were attempted with 20 seconds between jumps. The difference between standing reach height and each VJ height was calculated and then converted to centimeters because the Vertec measures VJ in inches. Most studies using the Vertec to measure VJ have used 3 attempts (1,13,18); however, subjects in this study were given 5 attempts and the best 3 VJ heights were averaged. This procedure was used because subjects were novices to VJ testing and the Vertec.

The starting point for the kicking test was marked on the midpoint of the end zone line and a regulation soccer goal was centered on the field, 70 yards away. An 8-yard wide target lane, the width of a soccer goal, was then marked from the starting line to the goal posts. All kicks landing outside the target lane were disqualified. Adidas Tango Supreme (size 5) (Adidas USA, Portland, Oregon, USA) soccer balls were used for all kick tests.

Subjects warmed up for the kick test by passing the ball to a partner 10 times at distances of 10, 20, and 30 yards. They then performed 1 practice kick consisting of a 1-step approach and kick toward the goal. Each subject then attempted 10 kicks with a 30-second rest between kicks. Kicking distance was measured in meters from the point where the ball was kicked to the point of initial contact with the ground. The best 5 kicks were averaged and used for data analysis.

General soccer training consisted of 3 soccer practices and 2 to 3 games per week. Practice typically lasted for 1.5 to 2 hours and focused on soccer-specific skills: dribbling, throw-ins, passing, tackling/defense, trapping/control, and heading. However, this was not controlled by the investigators and was left up to the individual coaches.

The weekly plyometric training consisted of various types of jumps, hops, skips, footwork, and sprint drills. Training in the first 6 weeks included single-leg forward hops over 6-inch cones, double-leg hops over 10-inch hurdles, lateral hops over 10-inch hurdles, and lateral shuffles over a 12-inch box. Training in the final 6 weeks included 10-inch box jump-ups, 10-inch depth jumps, and cutting drills. The depth jumps were considered low-intensity because of the height of 10 inches. The specific plyometric training program is detailed in Table 1.

TABLE 1. Plyometric regimen used with training group.

	Exercise	Direction	Sets	Repetitions
Pre-test				
Weeks 1 to 6	6-inch single-leg cone hops, 10 cones	Forward	1	5
	10-inch hurdle hops, 10 hurdles	Forward	1	5
	10-inch hurdle hops, 10 hurdles	Lateral	1	6
	12-inch box shuffles over a single box	Lateral	4	20
7-week test				
Weeks 8 to 13	10-inch box jump ups	Forward	3	10
	10-inch box depth jumps	Forward	3	10
	Cutting drills around cones (5 yards apart with 5-yard stagger)	Diagonal	2	10
14-week test				

Statistical Analyses

The design of the study was a 2 [Training (Plyometric, Control)] $\times$ 3 [Test (pre-test, 7 weeks, 14 weeks)] mixed design. The dependent measures of interest were VJ and kicking distance. Data were analyzed using mixed-model analysis of variance (ANOVA). SPSS for Windows, Release 16.0.1 (Statistical Package for Social Science, Chicago, Illinois, USA), was used for the analysis. Significance was set at the 0.05 alpha level.

RESULTS

Kicking Distance

The Training $\times$ Test interaction was significant ($F(2,28) = 53.54$, $p < 0.001$) indicating inconsistent results between the control and plyometric groups across time. Simple main-effects analysis was used to determine the nature of the significant interaction. Repeated-measures ANOVAs revealed significant differences in kicking distance across time for both the control group ($F(2,10) = 9.82$, $p = 0.004$, $\eta^2 = 0.663$) and the plyometric group ($F(2,10) = 65.69$, $p < 0.001$, $\eta^2 = 0.879$). Bonferroni pairwise comparisons of the control group's means across time (pre-test, 7 weeks, 14 weeks) revealed a significant decrease in kicking distance of 15.6% between the pre-test and 14 weeks ($p = 0.044$) and a significant decrease in kicking distance of 11.6% between 7 and 14 weeks ($p = 0.024$). Conversely, the plyometric group displayed significant increases in kicking distance: 11.5% from pre-test to 7 weeks ($p = 0.004$), 11.3% from 7 weeks to 14 weeks ($p < 0.001$), and 21.5% from pre-test to 14 weeks ($p < 0.001$) (Table 2).

The second phase of the simple main-effects analysis used independent t -tests to compare the control group to the plyometric group at each of the 3 test levels. The Bonferroni correction to alpha ($0.05/3 = 0.017$) was used to protect the type 1 error rate. No significant differences in kicking distance were found between the control and plyometric groups during the pre-test ($t(17) = 0.688$, $p = 0.500$, $\eta^2 = 0.027$, power = 0.10) or at 7 weeks ($t(16) = -1.665$, $p = 0.117$, $\eta^2 = 0.146$, power = 0.343). However, the plyometric group had significantly greater average kicking distance than the control group after 14 weeks ($t(17) = -6.332$, $p < 0.001$, $\eta^2 = 0.702$).

Vertical Jump

The Training $\times$ Test interaction was significant ($F(2,28) = 17.25$, $p < 0.001$, $\eta^2 = 0.552$), indicating inconsistent results within and between the control and plyometric groups across time. Simple main-effects analyses were used to determine the nature of the significant interaction. Repeated-measures ANOVAs revealed no significant differences in VJ height

TABLE 2. Average kicking distance in meters (mean $\pm$ SD) measured at the 3 test times.

Group	Test time		
	Pre-test	7 weeks	14 weeks
Plyometric ($n = 10$)	25.9 $\pm$ 2.6*†	29.3 $\pm$ 4.4‡	33.0 $\pm$ 3.7
Control ($n = 6$)	27.6 $\pm$ 2.5§	26.4 $\pm$ 3.8¶	23.3 $\pm$ 3.7

*Kicking distance increased 11.5% from pre-test to 7 weeks ($p = 0.004$).

†Kicking distance increased 21.5% from pre-test to 14 weeks ($p < 0.001$).

‡Kicking distance increased 11.3% from 7 weeks to 14 weeks ($p = 0.001$).

§Kicking distance decreased 15.6% from pre-test to 14 weeks ($p = 0.044$).

¶Kicking distance decreased 11.6% from 7 weeks to 14 weeks ($p = 0.024$).

TABLE 3. Average vertical jump distance in centimeters (mean $\pm$ SD) measured at the 3 test times.

Group	Test time		
	Pre-Test	7 weeks	14 weeks
Plyometric ($n = 10$)	39.6 $\pm$ 8.2* $\dagger$ cm	42.9 $\pm$ 6.8 $\ddagger$ cm	47.0 $\pm$ 8.1 cm
Control ($n = 6$)	39.4 $\pm$ 8.3	36.4 $\pm$ 6.8	36.8 $\pm$ 6.2

*Vertical jump height increased 8.3% from pre-test to 7 weeks ($p = 0.019$).

$\dagger$ Vertical jump height increased 18.6% from pre-test to 14 weeks ($p = 0.001$).

$\ddagger$ Vertical jump height increased 9.5% from 7 weeks to 14 weeks ($p = 0.003$).

across time (pre-test, 7 weeks, 14 weeks) in the control group ($F(2,10) = 2.13$, $p = 0.1697$, $\eta^2 = 0.299$, power = 0.336). The plyometric group displayed significant increases in VJ height across time ($F(2,10) = 30.46$, $p < 0.001$, $\eta^2 = 0.772$). Bonferroni pairwise comparisons revealed significant increases at each increment: 8.3% from pre-test to 7 weeks ($p = 0.019$), 9.5% from 7 weeks to 14 weeks ($p = 0.003$), and 18.6% from pre-test to 14 weeks ($p < 0.001$) (Table 3).

The second phase of the simple main-effects analysis compared the control group to the plyometric group at each of the 3 test levels. Independent t -tests and a Bonferroni-corrected alpha level of 0.017 were used in the analysis. No significant differences were found between the VJ heights in the control and plyometric groups during the pre-test ($p = 0.837$, $\eta^2 = 0.003$, power = 0.054) or at 7 weeks ($p = 0.108$, $\eta^2 = 0.154$, power = 0.360). However, the plyometric group had a significantly higher average VJ than the control group after 14 weeks ($p = 0.014$, $\eta^2 = 0.304$).

DISCUSSION

The present study is unique because it investigated low-frequency, low-intensity plyometric training with female adolescent soccer players. Myer et al. (26) tested the effect of a variety of low-intensity plyometric exercises with high school female athletes and found a significant increase in power. Siegler et al. (32) tested what was described as a high-intensity plyometric program using high school female soccer players and found a significant increase in VJ. These studies used a frequency of 3 days/week as compared to 1 day/week in the present study. Despite the lower frequency in the present study, the plyometric training group increased kicking distance 21.5% and VJ 18.6%, whereas the control group showed no improvement and actually declined in kicking distance performance as the season progressed. Robinson et al. (30) provided further evidence for lower-intensity plyometric exercises when they showed aquatic plyometric exercises provided the same performance enhancement as land plyometric exercises with significantly less muscle soreness.

Plyometric training has been shown to produce significant results in as few as 4 weeks (6). Luebbbers et al. (23) recommended 4- to 7-week training blocks and Nunez et al. (28) recommended 6- to 8-week training blocks for improvements in jumping ability. In the present study, significant changes did not take place until weeks 7 through 14. Similar results were noted by Moore et al. (25) using low-intensity (non-depth jump) plyometrics.

In their study, VJ decreased in the first 7 weeks before increasing 7% by week 14. The discrepancy in time to produce significant results may lie in the selected intensity and/or frequency of plyometric training. Most studies investigated the use of high-impact, advanced methods of plyometric training such as depth jumps 2 to 4 times per week (1,8,11,13,17,18,23), and LaChance (22) specifically recommended that plyometric training be completed on 2 days per week. This type of training may be contraindicated in adolescent athletes to protect the growth plates and skeletal development. However, some degree of low-impact plyometrics is part of normal adolescent activities such as skipping, hopping, and jumping. The prescribed training program used in the present study is no more physically demanding than traditional soccer training. Furthermore, the frequency of 1 day per week allowed adequate recovery time.

Results of the present study lend additional support to the notion that plyometric training increases VJ and therefore increases lower-body power. Many studies have shown a significant increase in VJ following a structured plyometric training regimen (1,8,11,13,14,17,18,23); however, the overwhelming majority of studies have been conducted using high-impact plyometric training, such as depth jumps. The results of the present study are important because we used a population that is not often studied. Minimal research addresses the effects of plyometric training with adolescent athletes. In addition, only males were used in the few studies that used subjects of a similar age to the present study. Brown et al. (8) and Hortobagyi et al. (18) investigated the effects of various plyometric training in young males who were freshmen and sophomores in high school and age 13.4 ± 0.11 years, respectively. The results for adolescent males were similar to the finding of the present study using females. Brown et al. (8) found gains of 11.2% in VJ with no arm movement and 12.5% with a double arm swing after training with depth jumps. Hortobagyi et al. (18) found significant improvement in both horizontal and VJ following 10 weeks of twice-weekly horizontal or vertical plyometric training.

This investigation also contributes normative values to the available body of knowledge for adolescent female athletes in

VJ and kicking distance. Prior to this study, there was virtually no information available concerning normative values for female adolescent soccer athletes. The subjects used in this study were able to kick distances ranging from 23.3 ± 3.7 to 33.0 ± 3.7 m and had VJ heights of 36.4 ± 6.8 to 47.0 ± 8.1 cm. Myer et al. (26) also assessed VJ in adolescent females but did not report specific values. However, from a figure it was determined that mean VJ heights were between 40 and 45 cm, which are similar to the results of the present study. Significant increases were shown in each study investigating the effects of plyometric training in adults, but the percentage gains were not as large. This may be caused by the difference in training and general fitness and development prior to plyometric training. Adults and college-age subjects have established a higher level of fitness prior to the beginning of testing than have adolescent athletes. This may explain why Brown et al. (8) and Hortobagyi et al. (18) both displayed such a considerable increase in VJ height in their young subjects.

PRACTICAL APPLICATIONS

Explosive power is important in most sports including soccer and should be a focus for strength and conditioning coaches. High-intensity plyometric training is often used by strength and conditioning coaches to increase power in mature athletes. However, because of the concern that high-intensity plyometric training will result in injury in skeletally immature athletes, this valuable training mode is often not used with adolescent athletes. The results of the present study demonstrate that significant improvements in activities that require power can be achieved with a once-weekly, low-intensity plyometric program. These results provide strength and conditioning coaches with a safe and effective alternative to high-intensity plyometric training. Based on these findings, to increase lower-body power resulting in increased VJ and kicking distance, strength and conditioning coaches should implement low-frequency, low-impact plyometric training programs with their adolescent athletes.

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